

Introduction to Systematic Design of Multi- Component Systems

SEAtS | 080172

Intended Learning Outcomes

By the end of this lecture, you will be able to:

- Define systematic and systemic design in the context of multi-component systems.
- Identify the differences between monolithic and distributed design approaches.
- Describe the key principles of a systematic design process.
- Explain the importance of interfaces and coordination in distributed system development.
- Evaluate tools and methods used to support the systematic design of complex systems.
- Apply a systematic design process to a simplified real-world engineering problem.

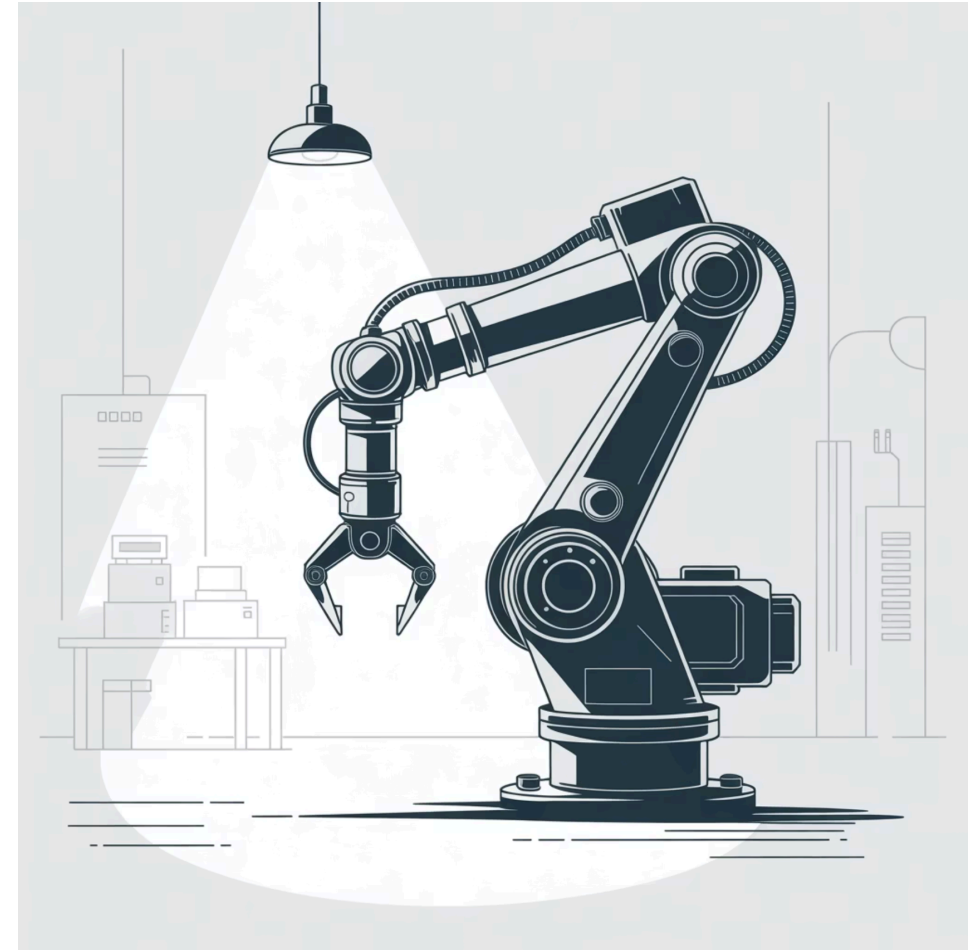
Problem statement

Modern products and systems are increasingly complex and interconnected, demanding **robust, adaptable and Scalable designs**; ad hoc approaches inherently increase risk.

Modern Engineering

Designing modern engineering systems often involves combining mechanical, electrical, and software components, each with its own functional logic and physical constraints.

A robot, for example, may contain a motion controller, sensors, actuators, communication links, and onboard processing. Such systems cannot be designed effectively without breaking the problem down, coordinating sub-systems, and validating their integration.



Systems Thinking

Systems thinking is a theoretical and analytical lens used to understand the behaviour of complex systems. It originates from systems theory (e.g. von Bertalanffy) and emphasises how parts of a system interact through feedback loops, interdependencies, and emergent behaviours, rather than analysing each part in isolation.

Key characteristics:

- Focus on the whole system, not just its parts
- Identifies feedback loops, delays, and non-linear cause-effect relationships
- Helps in understanding why systems behave as they do over time
- Often used for diagnosis, modelling, and scenario planning

Systems thinking underpins systemic design (systems thinking applied to design practice) and is applied specifically to AI systems in "Systems Thinking for AI Engineers".

Systems Thinking for AI Engineers

Systems thinking is a theoretical and analytical lens for understanding complex systems by focusing on interactions, feedback loops, interdependencies, and emergent behaviour — in contrast to reductionism, which analyses each part in isolation.

Applied to AI, it reframes AI systems as socio-technical ecosystems rather than isolated algorithms: an AI model's success depends on data flows, human and organisational context, ethics, and regulation, not just technical accuracy.

Key concepts include

- holism vs. reductionism,
- interconnectedness,
- system boundaries,
- reinforcing/balancing feedback loops,
- emergence,
- hierarchy and nested systems,
- delays and nonlinearity, and
- adaptation/self-organisation (e.g. federated learning).

Continued

Systemic design extends this into practice — designing AI ecosystems, not just algorithms, through participatory, stakeholder-inclusive methods using tools such as causal loop diagrams, giga-mapping, and stakeholder analysis matrices — and connects to distributed optimisation, where AI training pipelines (data preprocessing, training, inference, deployment) are decomposed into coordinated subsystems, mirroring distributed design of multi-component systems.

Systemic Design

A design approach that integrates systems thinking into the design process to address complex, interconnected, socio-technical challenges involving multiple stakeholders and dynamic relationships.

Key characteristics:

- Emphasises relationships, feedback loops, and emergent behaviours
- Integrates qualitative and quantitative analysis
- Focused on sense-making and co-creation in messy, real-world contexts
- Often applied to sustainability, healthcare, policy, education, and ecosystems

Systematic Design

Systematic Design is a structured approach that decomposes systems into interacting components, applying design principles to manage intricate challenges effectively.

1. **Structured:** Following repeatable processes with defined stages and checkpoints rather than intuitive approaches. This ensures consistency and quality across projects.
2. **Decomposable:** Breaking complex problems into manageable sub-problems that can be addressed independently, while maintaining awareness of interactions.
3. **Iterative:** Embracing feedback loops to continuously refine designs based on testing, simulation, and stakeholder input throughout the development lifecycle.

Systematic design is crucial for managing complexity, coordinating distributed teams, and ensuring that each subsystem contributes effectively to the global performance objectives. It is not simply a way to document a design process.

Question



The Elegance of the Paperclip

Its elegance lies in its systematic approach to a singular function. This humble office tool exemplifies several key principles of systematic design:

✓ Clear Functionality

Securely holds papers together without tearing or damaging them, fulfilling its core purpose flawlessly.

✓ Material Efficiency

Utilises a single, continuous piece of bent wire, minimising waste and complexity in its construction.

✓ Ease of Manufacturing

Its simple form allows for cost-effective mass production, making it widely accessible and affordable.

✓ Universal Adaptability

Recognised and effective worldwide, demonstrating a timeless and robust design solution for a common problem.

Real-World Complexity: EV

An EV is a highly integrated ecosystem of disparate components, each requiring meticulous design and seamless coordination. Which hinges on the harmonious interaction of these diverse systems.

Powertrain

Integrated electric motors, high-voltage batteries, and intricate power electronics for propulsion.

Software Systems

Advanced Battery Management Systems, autonomous driving, and sophisticated infotainment.

Chassis & Safety

Lightweight structures, advanced suspension, and comprehensive safety features protecting occupants.

User Experience

Intuitive digital interfaces, connectivity, and intelligent climate control for passenger comfort.

Question



Systematic Design Process Overview

The systematic design process can be represented in the following stages:

1. Inquiry – Understand system context, user needs, and constraints
2. Framing & Requirements Gathering – Define the scope, boundaries, and stakeholder expectations
3. System Architecture & Interface Definition – Decompose system into components; define interaction contracts
4. Component Design & Optimisation – Apply appropriate methods (e.g., FEA, simulation, surrogate modelling)
5. Integration & Testing – Assemble components, verify performance against system requirements
6. Feedback & Iteration – Use simulation and real-world data to refine models

Tools Supporting Systematic Design

To support these phases, engineers and designers use a range of tools:

1) Prediction / surrogate models

2) Optimization Algorithms

3) Simulation and Validation

4) System Mapping Tools

5) Prototyping and Implementation

Question



Understanding Multi-Component Systems

A multi-component system is composed of physically or functionally distinct elements that work together to deliver a complete functionality. Effective system design requires balancing these elements:

1. System-level requirements that address the overall purpose
2. Component-level specifications that drive individual elements
3. Interface definitions that ensure compatibility

Design Drivers

1. Performance metrics (speed, accuracy, efficiency)
2. Cost constraints (materials, manufacturing, lifecycle)
3. Modularity requirements (maintenance, upgradeability)
4. Time-to-market pressures

Challenges in Systematic Design of Multi-Component Systems

Challenge	Impact
Interface complexity	Integration failures, rework
Cross-disciplinary coordination	Misalignment, communication breakdown
Requirements instability	Design churn, rework
Scalability	Process overhead and design delays
Limited design exploration	Missed innovation opportunities
Tool integration	Inefficiencies, data loss
Change propagation	System-wide knock-on effects
Testing bottlenecks	Undetected errors, delayed feedback
Traceability management	High documentation cost
Human factors	Non-compliance, lack of buy-in

Interface complexity

A key challenge in multi-component systems is managing the interfaces between components. Interfaces define how data, energy, or force is exchanged between components and are critical to system functionality.

Mechanical Design

In mechanical design, interface stiffness matrices describe how forces and displacements are related across component boundaries.

Software Systems

In software systems, APIs define function calls and data structures.

Embedded Systems

In embedded systems, protocols govern timing and synchronization.

This allows each component to be optimized independently, However, such decomposition risks missing globally optimal solutions or producing physically infeasible designs.

Monolithic vs Distributed Design

Monolithic design

Monolithic design approaches attempt to optimize all design variables simultaneously in a single, tightly integrated process. This has the advantage of potentially achieving globally optimal solutions but suffers from scalability issues and coordination challenges in practice.

Distributed design

Distributed design optimization, on the other hand, decomposes the global design problem into subproblems, each associated with a component or subsystem. Coordination is then required to ensure that the subproblem solutions are compatible and converge to a system-level optimum.

This shift from global to local design parallels organisational design: individual teams can work on parts as long as shared protocols and interfaces are respected.

Question



Conclusion

Systematic design provides a structured and methodical way to manage the complexity inherent in large, multi-component systems.

It is both a mindset and a methodology, drawing from systems thinking, optimisation, and domain-specific engineering practices.

By integrating stakeholder needs, technical constraints, and adaptive feedback, systematic design becomes essential not just for building systems that work, but for building systems that can evolve.

Thank you

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